SPCNet: A Subpixel Convolution-Based Change Detection Network for Hyperspectral Images With Different Spatial Resolutions

Lifeng Wang, Liguo Wang, Heng Wang, Xiaoyi Wang, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*— The very high spectral resolution in hyperspectral images (HSIs) offers an opportunity to detect subtle land-cover changes. However, the availability of HSIs acquired from different platforms requires the development of change detection (CD) methods capable of processing HSIs with different spatial resolutions. In this article, we propose a general end-to-end subpixel convolution-based residual network (SPCNet) to accomplish the CD task between high spatial resolution (HR) and low spatial resolution (LR) HSIs. To effectively tackle the resolution matching issue, a super-resolution (SR) block with an efficient subpixel convolution layer is introduced to upscale the LR feature maps into HR maps. The subpixel convolution layer can fully explore the subpixel context information by learning an array of upscaling filters. Moreover, the designed SPC module is embedded into the LR branch to generate more discriminative representations. More importantly, the SPC module as a plug-and-play unit has the potential to be embedded into other baseline networks to enhance the feature learning capability. Experimental results on four HSI datasets demonstrate the effectiveness of the proposed SPCNet.**

***Index Terms*— Change detection (CD), deep learning, hyperspectral images (HSIs), multiscale images, remote sensing, residual network, subpixel convolution.**

I. Introduction

HYPERSPECTRAL images (HSIs) are valuable data sources to monitor global changes for Earth observation [1]. Change detection (CD) methods can recognize the differences in multitemporal HSIs acquired over the same geographical area at multiple times [2], [3]. They can be extensively exploited in many research activities, such as forestry and agriculture monitoring [4], [5], land sprawl dynamic analysis [6]–[8], and natural disaster damage assessment [9]–[11]. With the successful launch of satellites such as NASA Earth Observation-1 (EO-1), the Chinese Gaofen-5 satellites, and the Italian Space Agency PRISMA, the availability of HSIs at a global scale has significantly increased. Accordingly, CD in HSIs has received increasing attention in recent years.

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Over the past few decades, many conventional methods have been proposed to detect changes in bitemporal HSIs. In general, these methods can be divided into three main categories: algebra-based methods, transformation-based methods, and postclassification comparison methods. Typical algebra-based methods, such as image subtraction [12], image ratio [13], and change vector analysis (CVA) [14], are sensitive to radiation and noise, which compromises the CD accuracy. The representative transformation-based methods include iteratively reweighed multivariate alteration detection (IR-MAD) [15], slow feature analysis (SFA) [16], and principal component analysis (PCA) [17]. Although transformation-based methods can transform high-dimensional HSIs into other feature spaces and thus reduce the redundancy, they have a drawback in the need to select appropriate threshold values to detect land-cover changes. With regard to postclassification comparison methods [18], [19], the class maps generated by the classification of two images independently are compared pixel-by-pixel to obtain the changed regions [20]–[22]. A common drawback of postclassification comparison methods is that the CD result is directly affected by the propagation of classification errors on both images.

Recently, deep learning-based methods have been introduced for addressing the HSI task [23]. In [24], a Caps-TripleGAN framework is proposed by exploring the 1-D structure triple generative adversarial network (TripleGAN) for sample generation and integrating CapsNet for HSI classification. Wang *et al.* [25] proposed a unified multiscale learning framework, which contains a multiscale spatial–channel attention mechanism and a multiscale shuffle block to improve the problem of land-cover map distortion in the HSI classification. Wang *et al.* [26] proposed an SSA-SiamNet method to adaptively extract features from input patch pairs of HSIs for CD. In [27], a multilevel encoder–decoder attention network is proposed, which fuses the hierarchical features from all of the convolutional layers to detect changes in the HSIs. In [28], a three-direction spectral-spatial convolution neural network (CNN) (TDSSC) that contains a change tensor that decomposes along the spectral direction and two spatial directions is developed for HSI CD. In [29], the recurrent 3-D fully convolutional network (Re3FCN) merges the superiorities of a 3-D fully convolutional network and a convolutional long short-term memory. It is developed to extract joint spectral–spatial–temporal features. Wang *et al.* [30] proposed a general end-to-end two-dimensional-CNN framework (GETNET),

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which uses a mixed-affinity matrix as the input to detect binary changes in HSIs. Nonetheless, the abovementioned methods were developed for bitemporal HSIs with the same spatial resolution and do not sufficiently consider the use of subpixel context information.

In general, conventional and deep learning-based CD methods are applied to bitemporal HSIs acquired by the same sensors with the same spatial resolution. However, they are not adequate to address CD when HSIs acquired by different sensors with different spatial resolutions are considered [31]. For example, this case occurs if a high spatial resolution (HR) HSI of a certain region is available at time $T1$, but only a low spatial resolution (LR) HSI of the same area is available at time $T2$. In these cases, it is necessary to design CD methods that can effectively analyze bitemporal HSIs with different spatial resolutions. These methods should address two main challenges: spatial resolution matching and design of specific CD algorithms.

To address the spatial resolution matching issue, the most intuitive method is downsampling the HR HSI to make LR images or interpolating the LR HSI to the HR HSI to obtain two HSIs with the same spatial resolution. Then, common CD methods can be applied to detect changes [31]. However, on the one hand, the downsampling step leads to a lack of detailed spatial information. On the other hand, common interpolation methods, such as linear, bilinear, and bicubic, are not able to bring additional spatial information to solve the reconstruction problem. Therefore, it is necessary to develop information reconstruction strategies that make full use of subpixel information to detect land-cover changes in bitemporal HSIs with different spatial resolutions.

Different from the abovementioned interpolation methods, subpixel-based methods have been explored to accomplish the CD task for remote sensing images. For example, Liu *et al.* [32] proposed a super-resolution (SR)-based CD network that combines adversarial learning and stacked attention modules for CD on high-resolution images with different spatial resolutions. In [33], a novel approach that combines the subpixel mapping based on a backpropagation neural network and soft classification was proposed to address subpixel land-cover CD for multispectral images with different spatial resolutions. In [34], an iterative method to detect changes for multispectral images at different spatial scales was proposed. It involves endmember estimation, spectral unmixing, land-cover fraction CD, and SR land-cover mapping. However, when considering the problem of CD in bitemporal HSIs, to the best of our knowledge, the case of images with different spatial resolutions has seldom been considered.

Based on the abovementioned analysis, we propose a subpixel convolution-based residual network (SPCNet) for CD in bitemporal HSIs with different spatial resolutions. In greater detail, the main contributions of this article are given as follows.

1) A general end-to-end SPCNet framework is proposed to solve the CD problem in bitemporal HSIs with different spatial resolutions, which can fully extract the subpixel information from LR HSIs.
2) An SR block with a subpixel convolution operator, which learns an array of upscaling filters, is introduced to upscale the LR feature maps into HR images. It fully exploits the subpixel context information and generates more discriminative representations.
3) The SPC module is designed as a plug-and-play unit and has the potential to be embedded into other baseline networks to enhance the feature learning capability in other HSI processing tasks.

The remainder of this article is organized as follows. Section II describes the proposed SPCNet method in detail. Section III reports the experimental results of the proposed method. Section IV discusses further issues and draws the conclusions of this article.

II. Methodology

Fig. 1 shows an overview of the proposed end-to-end CD framework based on the SPCNet to detect land-cover changes in HSIs with different spatial resolutions. First, modified residual blocks are stacked to form an improved ResNet, which is the baseline to detect land-cover changes. Second, a pixel shuffling operator is embedded to form the SR block, which upscales the LR feature maps into HR maps. Third, the SPC module is designed to generate more discriminative subpixel-based representations from LR feature maps. Finally, the subpixel-based features of the LR HSI and the pixel-based features of the HR HSI are concatenated and given as input to the modified ResNet.

The four main parts of the proposed SPCNet method, i.e., the SR block, the SPC module, the Res block, and ResNet, as well as the loss function and the optimization strategy, are introduced in the following.

A. SR Block

The two HSIs with different spatial resolutions acquired at times $T1$ and $T2$ are represented as H_1^{HR} and H_2^{LR}, respectively. Let us assume that the same number b of spectral bands is associated with both H_1^{HR} and H_2^{LR}. The spatial sizes of H_1^{HR} and H_2^{LR} are $h_1 \times w_1$ and $h_2 \times w_2$, respectively, where $h_2 \leq h_1$ and $w_2 \leq w_1$. H_2^{LR} is interpolated as follows:

$$H_2'^{\mathrm{LR}} = \mathrm{up}\left(H_2^{\mathrm{LR}}\right) \tag{1}$$

where up$(\cdot)$ is the bicubic interpolation operator and the size of $H_2'^{\mathrm{LR}}$ is $h_1 \times w_1 \times b$. To make full use of the spatial context information, a set of patches from H_1^{HR} and $H_2'^{\mathrm{LR}}$ is constructed by combining a center pixel and its s surrounding pixels, thus producing patches I_1^{HR} and $I_2'^{\mathrm{LR}}$ with size $s \times s \times b$. To reduce the interference of redundant information caused by the interpolation strategy and the related processing, $I_2'^{\mathrm{LR}}$ is transformed as follows:

$$I_2^{\mathrm{LR}} = \mathrm{down}\left(I_2'^{\mathrm{LR}}\right) \tag{2}$$

where down$(\cdot)$ is the downsampling operation, which exploits the same strategy as that used in the interpolation operation to restore the original spectral information as much as possible. The patch size of I_2^{LR} is $d \times d \times b$ and $d \leq s$.

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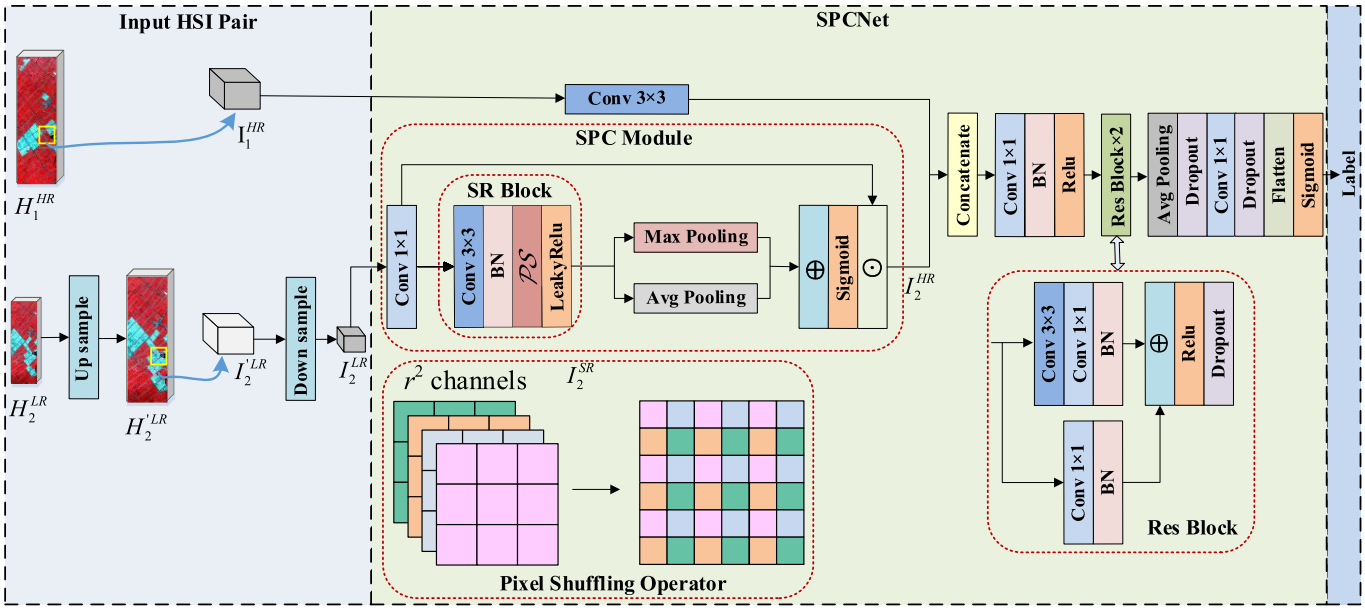


Fig. 1. Overview of the proposed SPCNet for CD on HSIs with different spatial resolutions.

To fully extract subpixel context information, a subpixel convolution layer is introduced to upscale the LR feature maps $\mathbf{I}_2^{\mathrm{LR}}$ into the HR output features into $\mathbf{I}_2^{\mathrm{SR}}$ [35], [35]. The subpixel convolution layer can be expressed as follows:

$$\mathbf{I}_2^{\prime\mathrm{SR}} = f^L\left(\mathbf{I}_2^{\mathrm{LR}}\right) = \mathcal{PS}\left(W_L * f^{L-1}\left(\mathbf{I}_2^{\mathrm{LR}}\right) + b_L\right) \tag{3}$$

where f^L denotes the L-th network layer and W_L and b_L are learnable network weights and biases, respectively. $\mathcal{PS}$ represents a periodic shuffling operator, which rearranges the elements of a $d \times d \times c \times r^2$ tensor to a tensor with $rd \times rd \times c$ shape. Moreover, r is an upscale factor that is set to 2 for experiments in Section III. Mathematically, the $\mathcal{PS}$ operation can be described as follows:

$$\mathcal{PS}(T)_{x,y,c_i} = T\lfloor x/r\rfloor, \lfloor y/r\rfloor, c_i \cdot r \cdot \mathrm{mod}(y,r) + c_i \cdot \mathrm{mod}(x,r). \tag{4}$$

These patterns are periodically activated during the convolution of the filter across the patch depending on different subpixel locations: $\mathrm{mod}(x,r)$ and $\mathrm{mod}(y,r)$, where x and y are the output pixel coordinates in the HR space. $\mathrm{mod}(\cdot)$ represents the modulo operator, which returns the remainder of dividing two numbers, and c_i is the ith filter channel.

In the proposed method, the SR block is designed to extract effective subpixel features and includes a convolutional layer, a batch normalization (BN) operation, a $\mathcal{PS}$ operation, and a LeakyReLU activation function. The specific arrangement of the SR block is shown in Fig. 1 and can be calculated as

$$\mathbf{I}_2^{\mathrm{SR}} = \sigma\left(\mathcal{PS}\left(g\left(f\left(\mathbf{I}_2^{\mathrm{LR}}\right)\right)\right)\right) \tag{5}$$

where $\sigma(\cdot)$ represents the LeakyReLU activation function and $g(\cdot)$ denotes the transform of the convolution and BN operations.

B. SPC Module

The proposed SPC module is arranged according to Fig. 1. First, the intermediate subpixel features are summarized as follows:

$$\mathbf{I}_2^{\mathrm{SR}} = \mathcal{F}\left(f\left(\mathbf{I}_2^{\mathrm{LR}}\right)\right) \tag{6}$$

where $f(\cdot)$ represents the convolution operation and $\mathcal{F}(\cdot)$ is the series of operations of the SR block. Then, the subpixel-based features $\mathbf{I}_2^{\mathrm{SR}}$ can be transformed as follows:

$$\mathbf{I}_2^{\mathrm{HR}} = f\left(\mathbf{I}_2^{\mathrm{LR}}\right) \cdot \sigma\left(\mathrm{MaxPool}\left(\mathbf{I}_2^{\mathrm{SR}}\right) + \mathrm{AvgPool}\left(\mathbf{I}_2^{\mathrm{SR}}\right)\right) \tag{7}$$

where $\mathrm{MaxPool}(\cdot)$ and $\mathrm{AvgPool}(\cdot)$ represent the max-pooling and avg-pooling operations, respectively, and "$\cdot$" denotes the elementwise multiplication. Note that this designed SPC module with auxiliary subpixel context information can be extended directly to the other frameworks.

C. Res Block and ResNet

ResNet has been introduced to robustly improve the performance of CNNs with a very deep architecture. It is an extension of the CNN with shortcut connections that promotes the propagation of gradients. Many residual units are stacked to form deep residual networks to avoid gradient extinction [36].

In our approach, the residual unit is modified specifically for the HSI CD task. Instead of the identity shortcut connection present in the traditional residual unit, in the designed residual block, a connection with 1×1 convolutional and BN operations is employed. More specifically, a dropout layer is adopted in the modified residual unit to avoid the overfitting problem. The improved residual unit can be described as follows:

$$h^{l+1} = G\left(g\left(f\left(h^l\right)\right)\right) + g\left(h^l\right) \tag{8}$$

where h^l and h^{l+1} are the input and output features of the lth unit, respectively, and $G(\cdot)$ represents the ReLU activation function and dropout operation.

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As shown in Fig. 1, the subpixel-based features of the LR branch and the pixel-based features of the HR branch are concatenated to be given as input to the designed ResNet. In this study, the CD problem is modeled as a binary classification task. Thus, a fully connected layer, including two filters and a sigmoid activation function, is adopted as the end of the proposed architecture. It provides as outputs a binary label that indicates whether the pixel has changed or not.

D. Loss Function and Optimization

An appropriate loss function is crucial to optimizing the designed network during model training. The cross-entropy function can measure the dissimilarity between the ground-truth labels and the predicted probabilities. Therefore, binary cross entropy is commonly utilized for model training in binary classification [37]. Accordingly, the binary cross entropy is adopted as the loss function of the proposed SPCNet method. It is defined as follows:

$$\text{Loss}_{\text{BCE}} = -\frac{1}{M}\sum_{i=1}^{M}\left[y_i \times \log\left(\hat{y}_i\right) + (1-y_i) \times \log\left(1-\hat{y}_i\right)\right] \tag{9}$$

where M represents the number of patch pairs in a mini-batch during the model training and y_i and $\hat{y}_i$ denote the ground-truth label and the predicted probability for the ith patch pair, respectively. The labels of the unchanged and changed patch pairs are set to $y_i = 0$ and $y_i = 1$, respectively. Stochastic gradient descent and backpropagation are applied to update the parameters in the designed network.

III. Experiments

A. Description of Datasets

To assess the performance of the proposed SPCNet method, four HSI datasets were used in the experiments: Farmland, River, Hermiston, and Bay Area. The first three datasets were acquired by using the Hyperion sensor onboard the EO-1 satellite. The Hyperion sensor covers wavelengths that range from 0.4 to 2.5 μm with 242 spectral bands. It provides HSIs with a spectral resolution of approximately 10 nm and a spatial resolution of approximately 30 m. The Bay Area dataset was acquired by an Airborne Visible Infrared Imaging Spectrometer (AVIRIS) from NASA's Jet Propulsion Laboratory (JPL). AVIRIS covers the spectral range from 0.4 to 2.5 μm with 224 spectral bands. The AVIRIS HSIs have a spatial resolution of 20 m and a spectral resolution of 10 nm.

The first dataset, named Farmland, covers a farmland area in Yancheng, Jiangsu Province, China. The bitemporal HSIs were acquired on May 3, 2006, and April 23, 2007. The spatial extent is 450 $\times$ 140 pixels, and 155 bands are selected for CD after removing noisy bands. Crop rotation represents the main changes in this dataset.

The second dataset, named River, contains a river region in Jiangsu Province, China. The bitemporal HSIs were acquired on May 3, 2013, and December 31, 2013. The spatial size is 463 $\times$ 241 pixels, and 198 bands are considered after the removal of noisy bands. The changes in the dataset are mainly related to the removal of sediment from the river.

The third dataset, named Hermiston, encompasses an irrigated agricultural field in Hermiston City, Umatilla Country, OR, USA. The pair of HSIs was acquired on May 1, 2004 and May 8, 2007. The images have a size of 307 $\times$ 241 pixels and contain 154 bands after noisy band removal. The changes in this dataset are caused mainly by the regulation of irrigation areas.

The fourth dataset, named Bay Area, covers an area in the San Francisco Bay Area, CA, USA. The bitemporal HSIs were acquired in 2013 and 2015. The image size is 600 $\times$ 500 pixels and the number of spectral bands is 224.

The bitemporal HSIs have the same spatial resolution in the above four datasets. However, bitemporal HSIs with different spatial resolutions are needed for subsequent experiments. Therefore, we first downsample the original HR image in $T2$ to simulate the corresponding LR version of the $T2$ image. The scale factor is the spatial resolution ratio between the LR and HR image pairs. The original HR $T1$ image and the corresponding LR $T2$ image form the multiresolution dataset. Examples of true-color compositions of three spectral channels of images with different spatial resolutions for the four datasets are shown in Fig. 2. The use of these downsampled images has some limitations but allows us to accurately assess the performance of the proposed SPCNet and compare it with the performances of other techniques.

B. Experimental Setup

The optimizer is based on the root-mean-square propagation (RMSprop). The number of total epochs is 200, and the initial learning rate is set to 0.001. The batch size is 64 for the Farmland and Bay Area datasets and 128 for the River and Hermiston datasets. The learning rate decays by a factor of 10 for every 50 epochs. An NVIDIA GeForce RTX3090 GPU was employed to conduct the experiments. The proposed SPCNet was implemented on the TensorFlow and Keras framework.

For a fair comparison, we randomly select 5% of the samples from the Farmland, River, and Hermiston datasets and 1% of the samples from the ground references of the Bay Area dataset as training sets. The remaining samples are used as test sets. In the experiments, all of the input data were standardized to zero mean values with unit variance, and ten independent tests were performed to avoid biased estimation.

To quantitatively assess the detection performance, we adopted the overall accuracy (OA), kappa coefficient (Kappa), precision (Pr), recall (Re), and F1-score (F1) as evaluation metrics [26].

C. Parameter Setting

In the proposed SPCNet method, several hyperparameters can affect the training model and, further, the CD results. They include the kernel numbers of convolutional filters, the patch size, the training sample proportion, and the scale factor. In this section, these hyperparameters are analyzed in detail.

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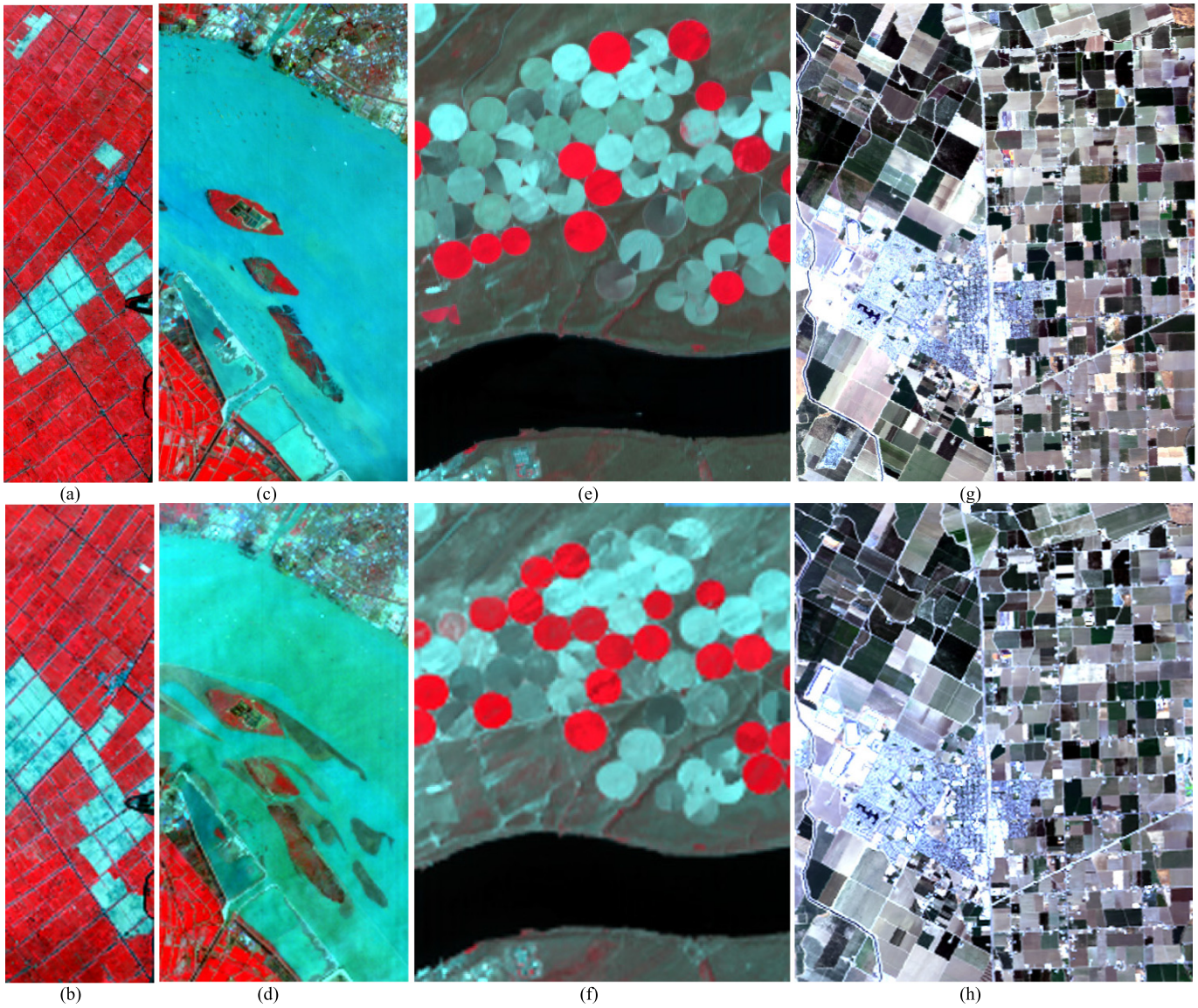


Fig. 2. Four HSI datasets used in the experiments (bands 33, 22, and 11 as RGB). (a) HR Farmland image at time $T1$. (b) LR Farmland image at time $T2$. (c) HR River image at time $T1$. (d) LR River image at time $T2$. (e) HR Hermiston image at time $T1$. (f) LR Hermiston image at time $T2$. (g) HR Bay Area image at time $T1$. (h) LR Bay Area image at time $T2$. The scale factor of the LR image is set to 2.

The impact of a certain parameter was explored by fixing the values of the other parameters.

1) Impact of the Kernel Numbers: The kernel numbers of the convolutional filters are related to the representation capacity and the computational cost of the SPCNet method. In the proposed SPCNet, the kernel number in each convolutional layer is the same except for the SR block. Different kernel numbers (i.e., 4, 8, 16, 24, 32, and 64) were examined in preliminary experiments. As shown in Fig. 3(a), as the kernel numbers increase from 4 to 32 and the OA values also increase in all four datasets. Compared with 32 kernels, 64 kernels increase the model complexity, but the accuracy is not significantly improved. Thus, the kernel number is fixed to 32 for all four datasets.

2) Impact of the Patch Size: The size of the input patch pair is related to the amount of spatial information surrounding the center pixel. To evaluate the influence of different patch sizes on the SPCNet, we considered a set of four patch sizes: 5×5, 7×7, 9×9, and 11×11. As shown in Fig. 3(b), as the patch size increases, the OA value decreases gradually in all of the datasets except for the Bay Area dataset. In our study, the patch size was set to 5×5 in all datasets.

3) Impact of the Batch Size: To explore the influence of the patch size on the proposed SPCNet, a set of batch sizes (32, 64, 128, 256, and 512) were examined. From Fig. 3(c), one can observe that the OA values do not change significantly under different batch sizes, which suggests that the proposed SPCNet is robust to the batch size. The optimal batch size is 64 for the Farmland and Bay Area datasets and 128 for the River and Hermiston datasets.

4) Impact of the Proportion of Training Samples: To demonstrate the generality of the proposed SPCNet method, we examined different proportions of training samples (0.1%, 0.5%, 1%, 5%, 10%, and 15%) for ResNet and SPCNet in all four datasets. As shown in Fig. 4, the OA values of ResNet and SPCNet increased by increasing the proportion

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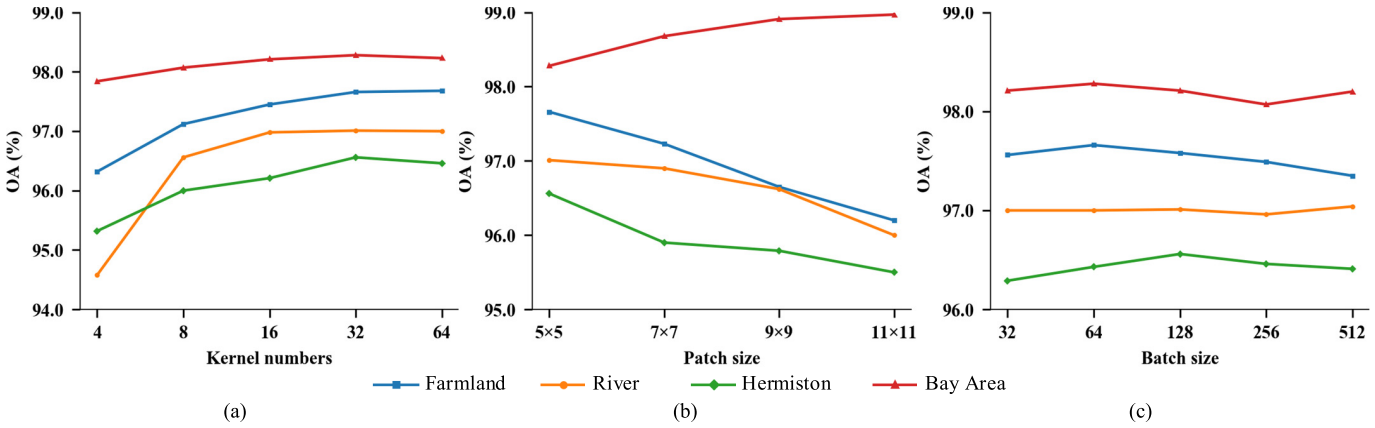


Fig. 3. OA values obtained by the SPCNet with different hyperparameter values for the four datasets: (a) kernel numbers, (b) patch size, and (c) batch size.

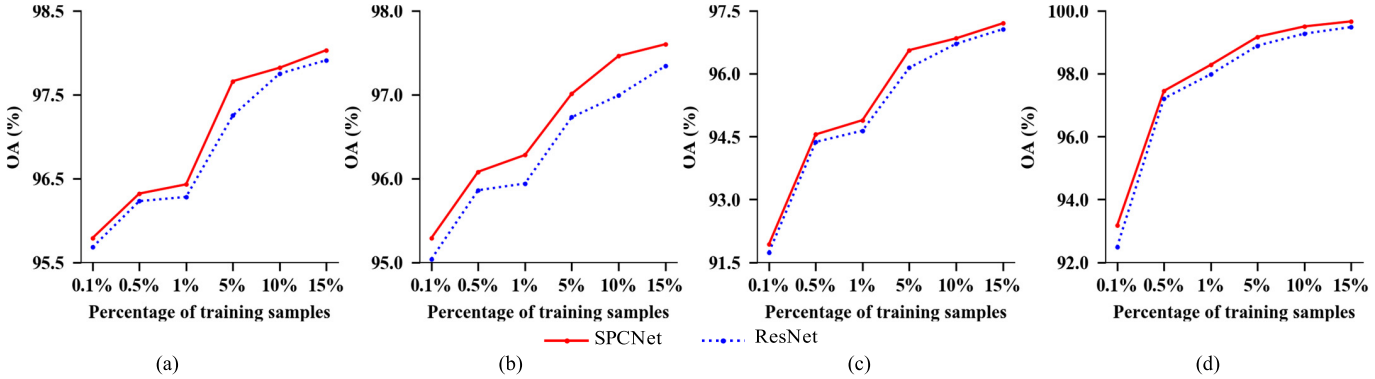


Fig. 4. OA values obtained by ResNet and SPCNet with different proportions of training samples for (a) Farmland dataset, (b) River dataset, (c) Hermiston dataset, and (d) Bay Area dataset.

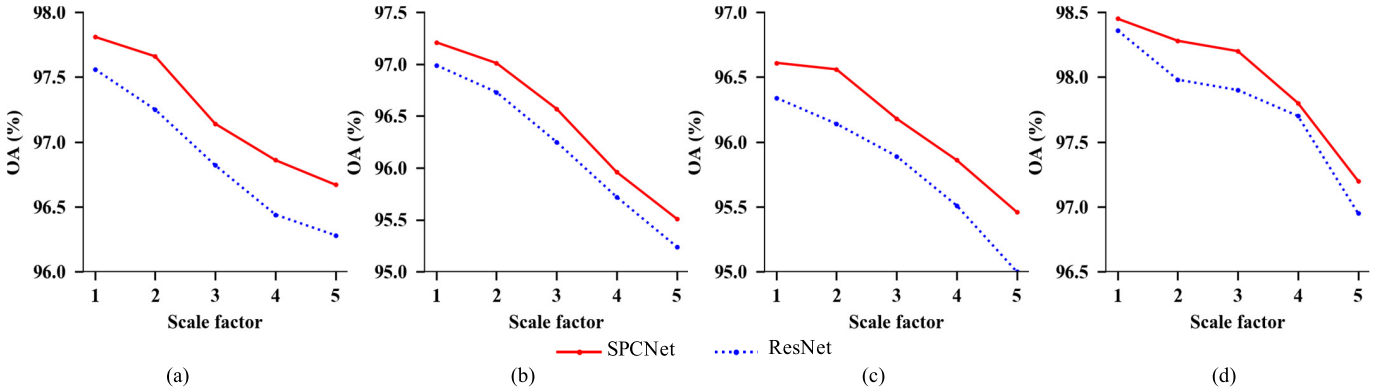


Fig. 5. OA values obtained by ResNet and SPCNet with different scale factors for (a) Farmland dataset, (b) River dataset, (c) Hermiston dataset, and (d) Bay Area dataset.

from 0.1% to 15% for the four datasets. In general, the proposed SPCNet method improves the OA value compared with ResNet.

5) Impact of the Scale Factor: To evaluate the effectiveness of the proposed SPCNet method, we considered a set of scale factors (1, 2, 3, 4, and 5) between the spatial resolutions of $T1$ and $T2$ images. From Fig. 5, one can see that as the scale factor increases, due to the decrease in the spatial resolution of the HSI at $T2$ time, the OA value decreases gradually in all datasets. As expected, the proposed SPCNet method is effective in detecting changes from bitemporal HSIs with the same spatial resolution (the scale factor is set to 1).

D. Comparison With Other Methods

To verify the effectiveness of the proposed SPCNet method, we compare it with 11 benchmark methods, which include CVA [38], SVM [39], SiamNet and SSA-SiamNet [26], GETNET [30], Sub-2DCNN [40], Con-2DCNN [40], Con-3DCNN [41], Sub-RSSAN and Con-RSSAN [42], and ResNet. In SVM, the radial basis function kernel was selected in the four datasets. The input images for the Sub-2DCNN and the Sub-RSSAN are the result of subtracting the HR image and the interpolated LR image. For Con-based methods (i.e., SVM, Con-2DCNN, Con-3DCNN, Con-RSSAN, and ResNet), the HR image and the interpolated LR image are concatenated and

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TABLE I

ACCURACY OF DIFFERENT CD METHODS ON THE FOUR CONSIDERED DATASETS (BOLD VALUES INDICATE THE HIGHEST ACCURACIES FOR EACH VARIABLE AND EACH DATASET)

Dataset	Method	Metrics				
		OA (%)	Kappa×100	F1 (%)	Pr (%)	Re (%)
Farmland	CVA	96.18	90.83	93.54	91.78	95.37
	SVM	97.73	94.49	96.09	96.34	95.84
	SiamNet	96.98±0.29	92.65±0.76	94.77±0.56	95.34±0.37	94.20±1.39
	SSA-SiamNet	97.04±0.36	92.80±0.68	94.87±0.49	95.45±0.24	94.30±0.76
	GETNET	97.56±0.16	94.05±0.42	95.77±0.62	96.38±0.44	95.16±0.51
	Sub-2DCNN	**97.74±0.04**	**94.51±0.09**	**96.10±0.14**	96.07±0.22	**96.14±0.15**
	Con-2DCNN	97.58±0.06	94.13±0.14	95.83±0.10	95.84±0.19	95.83±0.13
	Con-3DCNN	97.59±0.08	94.13±0.25	95.82±0.37	96.54±0.31	95.11±0.23
	Sub-RSSAN	97.44±0.04	93.78±0.08	95.58±0.05	95.64±0.08	95.53±0.06
	Con-RSSAN	97.50±0.05	93.92±0.12	95.67±0.09	96.28±0.12	95.08±0.13
	ResNet	97.25±0.07	93.31±0.18	95.24±0.13	95.60±0.21	94.89±0.27
	Proposed SPCNet	97.66±0.05	94.30±0.14	95.94±0.09	**96.55±0.22**	95.34±0.31
River	CVA	82.96	39.79	47.45	32.42	88.52
	SVM	96.31	73.81	75.77	88.15	66.44
	SiamNet	96.57±0.25	76.86±0.19	78.72±0.25	85.06±0.51	73.26±0.41
	SSA-SiamNet	96.70±0.31	78.88±0.25	80.68±0.43	82.17±0.57	79.24±0.62
	GETNET	96.50±0.26	76.68±0.61	78.57±0.53	84.05±0.49	73.76±0.95
	Sub-2DCNN	96.65±0.04	78.50±0.30	80.33±0.28	82.01±0.11	78.72±0.50
	Con-2DCNN	96.61±0.04	78.48±0.17	80.34±0.15	81.08±0.41	79.61±0.14
	Con-3DCNN	96.57±0.09	77.26±0.59	79.12±0.62	83.93±0.72	74.84±0.89
	Sub-RSSAN	96.88±0.04	79.88±0.24	81.58±0.22	83.61±0.56	79.66±0.52
	Con-RSSAN	96.68±0.04	78.37±0.20	80.18±0.18	83.30±0.44	77.29±0.33
	ResNet	96.73±0.04	79.14±0.44	80.93±0.42	82.01±0.59	**79.88±1.25**
	Proposed SPCNet	**97.01±0.09**	**80.29±0.93**	**81.91±0.89**	**86.43±1.62**	77.93±0.25
Hermiston	CVA	92.94	77.48	81.70	98.22	69.94
	SVM	95.65	87.04	89.79	95.35	84.84
	SiamNet	95.90±0.24	88.25±0.56	90.90±0.32	90.59±0.66	91.21±0.35
	SSA-SiamNet	96.22±0.34	89.02±0.43	91.45±0.43	93.18±0.71	89.77±0.52
	GETNET	95.58±0.27	87.50±0.41	90.37±028	88.75±0.96	92.05±0.89
	Sub-2DCNN	96.17±0.04	88.89±0.12	91.35±0.09	92.81±0.30	89.94±0.21
	Con-2DCNN	96.48±0.45	89.93±0.13	92.20±0.11	92.22±0.21	**92.17±0.30**
	Con-3DCNN	96.10±0.41	88.68±0.58	91.19±0.32	92.87±0.51	89.56±0.46
	Sub-RSSAN	95.67±0.04	87.48±0.10	90.26±0.08	91.53±0.20	89.03±0.18
	Con-RSSAN	95.85±0.06	88.00±0.18	90.67±0.15	91.97±0.13	89.40±0.31
	ResNet	96.14±0.05	88.86±0.26	91.34±0.19	92.23±0.48	90.47±0.46
	Proposed SPCNet	**96.56±0.04**	**90.05±0.13**	**92.26±0.10**	**93.51±0.44**	91.06±0.05
Bay Area	CVA	82.15	64.81	80.81	94.99	70.32
	SVM	94.78	89.53	95.07	96.02	94.14
	SiamNet	97.53±0.74	95.03±0.81	97.71±0.65	96.55±0.63	**98.90±0.95**
	SSA-SiamNet	97.62±0.69	95.21±0.96	97.76±0.57	98.08±0.71	97.45±0.83
	GETNET	93.85±0.56	87.65±0.75	94.26±0.43	94.18±0.53	94.33±0.76
	Sub-2DCNN	96.95±0.18	93.87±0.22	97.12±0.19	97.97±0.21	96.28±0.27
	Con-2DCNN	97.58±0.13	95.13±0.25	97.72±0.12	98.13±0.17	97.32±0.12
	Con-3DCNN	97.08±0.28	94.15±0.55	97.24±0.35	98.32±0.61	96.19±0.43
	Sub-RSSAN	96.91±0.27	93.80±0.42	97.08±0.42	98.11±0.51	96.07±0.59
	Con-RSSAN	97.31±0.33	94.60±0.64	97.45±0.38	98.62±0.47	96.32±0.75
	ResNet	97.98±0.48	95.96±0.97	98.11±0.45	98.60±0.45	97.62±0.58
	Proposed SPCNet	**98.28±0.12**	**96.55±0.24**	**98.39±0.11**	**98.92±0.24**	97.85±0.10

given as input to the model. For deep learning-based methods, the specific parameter settings refer to [26].

1) Quantitative Comparison: Table I reports the CD accuracy of different methods on the four datasets. By analyzing the table, we can derive the four following observations.

First, the CVA method provides the smallest accuracy for the four datasets, especially for the River and Bay Area datasets. Compared with supervised approaches, CVA is a classic unsupervised method. It does not require any training samples that prevent us from accurately distinguishing various changes.

Second, the accuracies of deep learning-based methods are superior to that of the SVM method on all datasets except for the Farmland dataset. The reason is that different from the SVM that exploits the spectral information, deep learning-based methods can make full use of the

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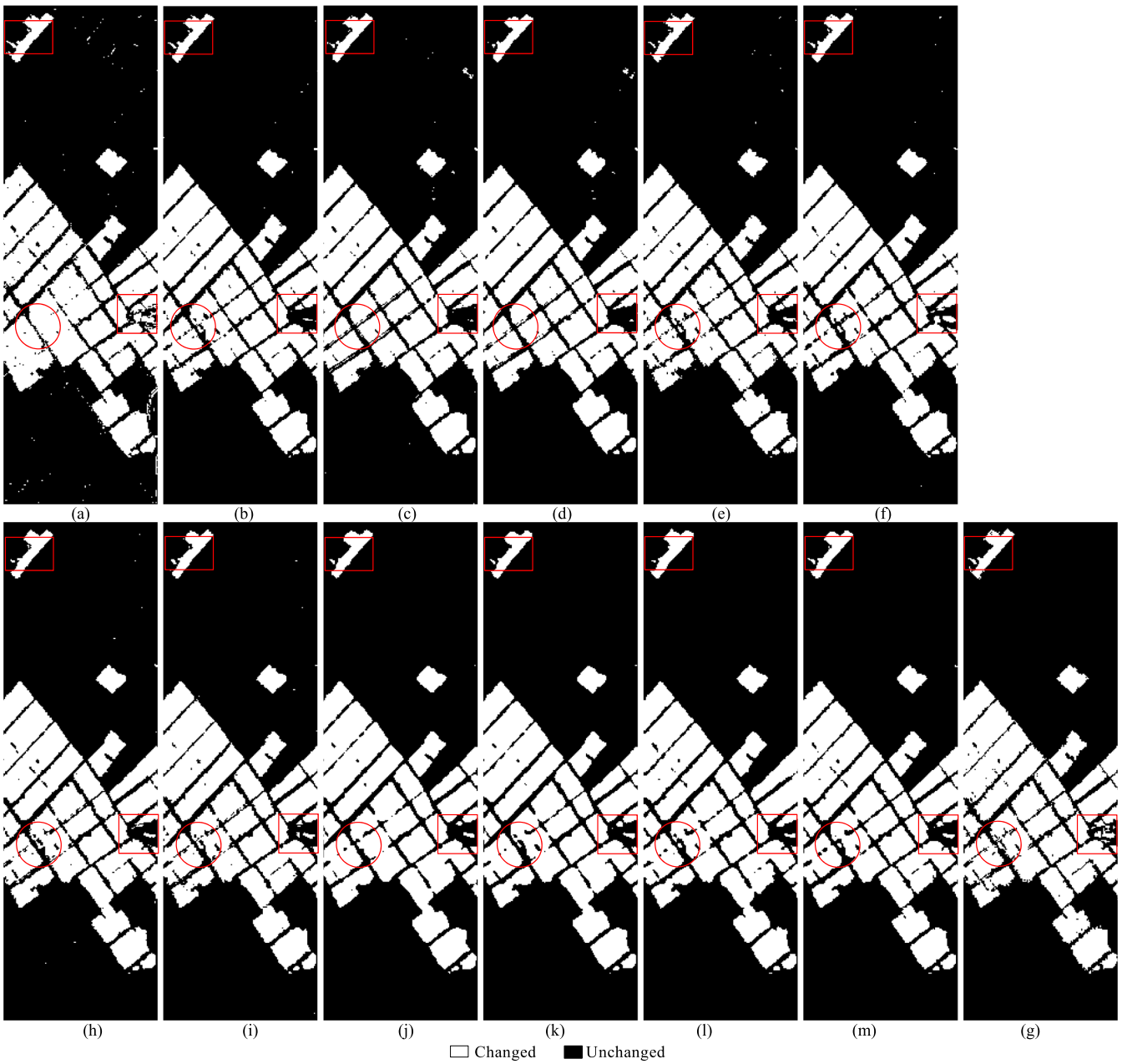


Fig. 6. CD maps produced by different methods on the Farmland dataset: (a) CVA, (b) SVM, (c) SiamNet, (d) SSA-SiamNet, (e) GETNET, (f) Sub-2DCNN, (g) Con-2DCNN, (h) Con-3DCNN, (i) Sub-RSSAN, (j) Con-RSSAN, (k) ResNet, (l) Proposed SPCNet, and (m) Ground-reference map.

spatial–spectral context information and generate more discriminative features. This process facilitates the detection of land-cover changes.

Third, compared with the SiamNet and GETNET methods, other deep learning methods provide metric values with lower standard deviations and are more centralized around the means. This finding indicates the stability of the algorithm.

Finally, the proposed SPCNet enhances the performance of ResNet on all datasets. More precisely, the increases in OAs for the Farmland, River, Hermiston, and Bay Area datasets are 0.41%, 0.28%, 0.42%, and 0.30%, respectively. The reason for these improvements is that the proposed SPCNet extracts more discriminative features through the subpixel convolution operation, which improves the CD accuracy.

2) Qualitative Comparison: Ground-reference maps for the four datasets are shown in Figs. 6(m)–9(m), where the white, black, and gray parts describe changed, unchanged, and unknown pixels, respectively. Moreover, Figs. 6–9 show the CD maps obtained by different methods on the four datasets for a comparative qualitative comparison. Areas that have the main differences in the maps are indicated by red squares and circles. Overall, the CVA maps contain more noise than those of the other supervised methods, which indicates the importance of the training information in detecting changes. Generally, compared with the SVM method, the CD maps of deep learning-based techniques are much more similar to the ground-reference map, which confirms the effectiveness of using the spectral–spatial context information. Furthermore,

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TABLE II

COMPUTATIONAL COSTS OF DIFFERENT METHODS ON THE FOUR CONSIDERED DATASETS (BOLD VALUES INDICATE THE LOWEST COMPUTATIONAL COST RESULTS FOR EACH VARIABLE AND EACH DATASET)

Dataset		Method									
		SiamNet	SSA-SiamNet	GETNET	Sub-2DCNN	Con-2DCNN	Con-3DCNN	Sub-RSSAN	Con-RSSAN	ResNet	SPCNet
Farmland	Parameters(K)	87.84	88.60	93420	55.43	73.13	155.07	49.25	103.05	**34.47**	80.42
	FLOPs (M)	**39.36**	39.60	55080	52.81	104.24	1130	41.25	83.63	88.98	90.88
River	Parameters (K)	106.42	107.18	154180	48.94	91.71	159.19	63.65	135.14	**37.22**	94.18
	FLOPs (M)	134.15	134.80	90910	100.13	132.77	1470	**52.87**	107.72	186.75	213.47
Hermiston	Parameters (K)	110.15	111.52	92180	55.14	99.49	72.07	37.17	79.53	**34.40**	80.10
	FLOPs (M)	104.95	105.60	54350	52.48	103.57	1120	**40.99**	83.09	177.74	181.03
Bay Area	Parameters (K)	166.08	167.32	198270	75.30	139.81	277.03	72.68	156.25	**38.88**	102.50
	FLOPs (M)	151.40	152.05	116910	75.71	150.02	1670	**59.98**	122.72	96.04	116.32

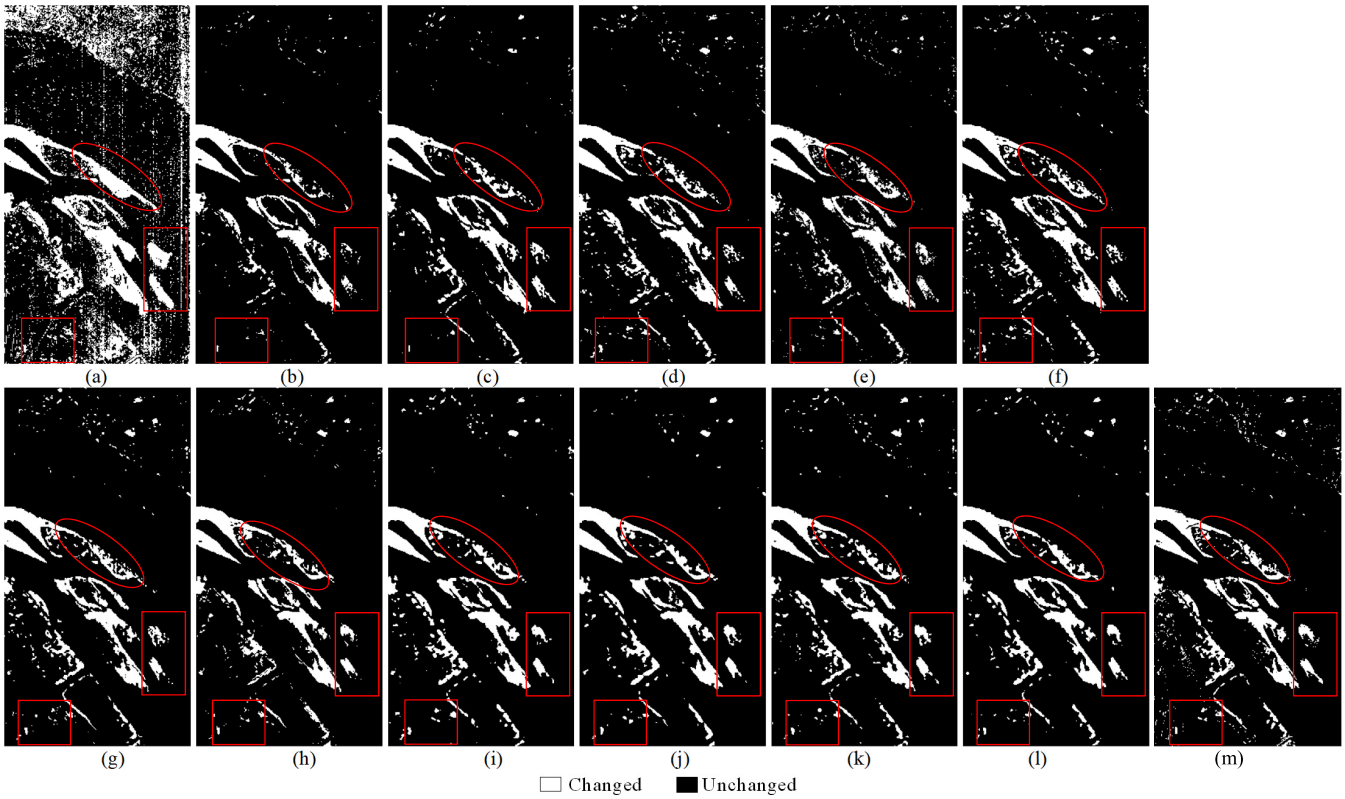


Fig. 7. CD maps produced by different methods on the River dataset: (a) CVA, (b) SVM, (c) SiamNet, (d) SSA-SiamNet, (e) GETNET, (f) Sub-2DCNN, (g) Con-2DCNN, (h) Con-3DCNN, (i) Sub-RSSAN, (j) Con-RSSAN, (k) ResNet, (l) Proposed SPCNet, and (m) Ground-reference map.

the CD map produced by the proposed SPCNet method is the closest to the ground-reference maps on all datasets. This aspect is consistent with the quantitative results in Table I.

3) Computational Cost: A number of trainable parameters and floating-point operations (FLOPs) are used to describe the computational cost of the deep learning-based methods. Table II lists the computational costs of ten deep learning-based methods. Obviously, GETNET requires the largest number of parameters and FLOPs among the nine methods for the four datasets. The reason is that the input data of GETNET are large patches with size $c \times c \times 1$. Overall, the modified ResNet involves the lowest number of parameters and FLOPs among all methods, which indicates the superiority of the improved ResNet as a baseline. Furthermore, the proposed SPCNet, which improves the CD accuracy, shows an acceptable slight increase in the model complexity.

E. Ablation Analysis

To verify the effectiveness of the proposed SPCNet method, we implemented an ablation study in which the hyperparameters were set according to Section III-C.

1) Analysis With and Without Downsampling Strategies: To investigate the importance of using the downsampling strategy in the proposed SPCNet method, we carried out experiments without using it. To ensure the feature size matching of the method without the downsampling strategy, a convolution layer with kernel size 3×3 is used at the $T2$ branch. As shown in Fig. 10, the CD results of the SPCNet with and without the downsampling strategy are relatively similar for the Farmland and River datasets. In contrast, the accuracy of the proposed SPCNet with the downsampling strategy is superior to that without it for the Hermiston and Bay Area datasets. From Table III, one can see that the computational costs in terms of the training parameters and

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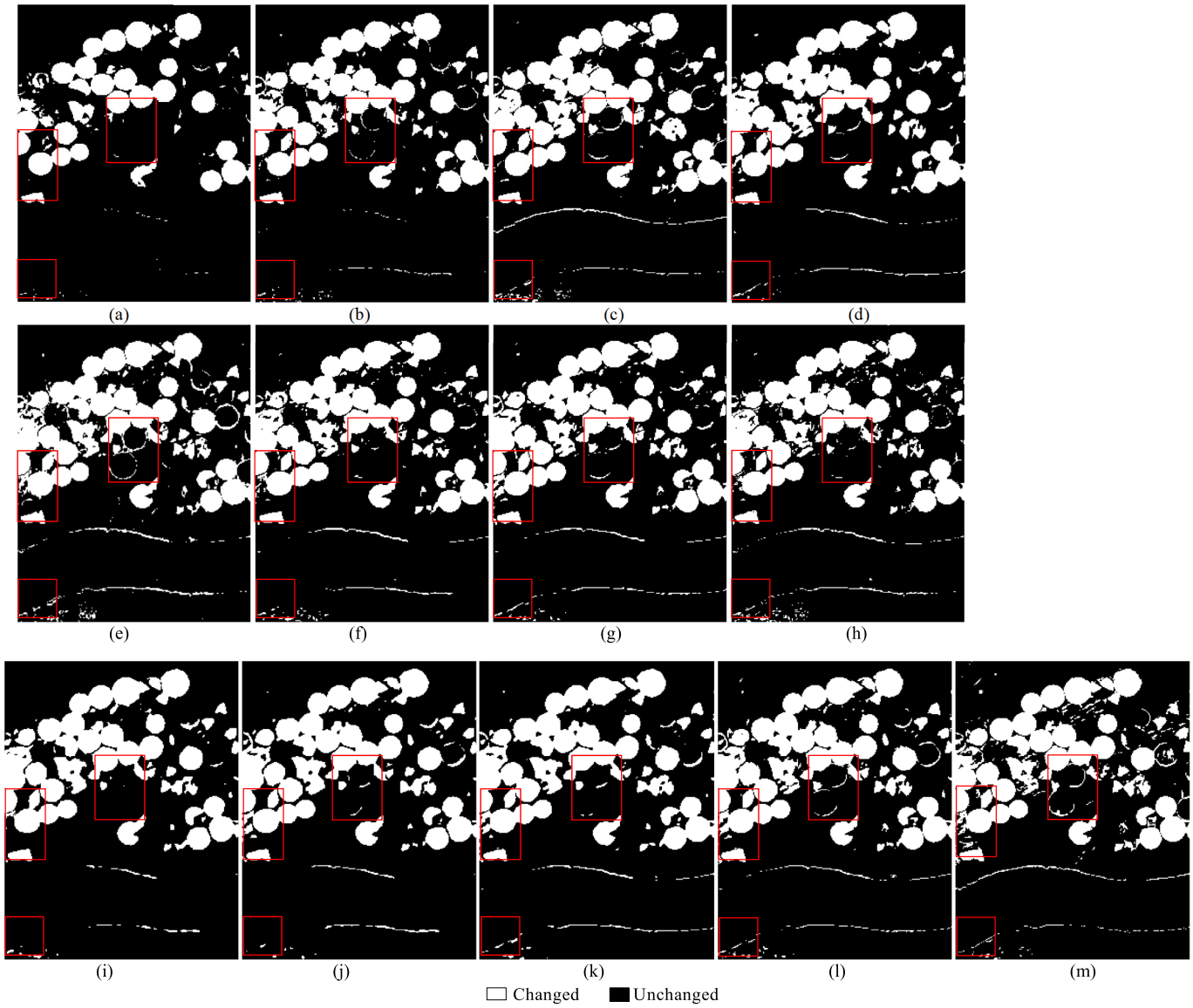


Fig. 8. CD maps produced by different methods on the Hermiston dataset: (a) CVA, (b) SVM, (c) SiamNet, (d) SSA-SiamNet, (e) GETNET, (f) Sub-2DCNN, (g) Con-2DCNN, (h) Con-3DCNN, (i) Sub-RSSAN, (j) Con-RSSAN, (k) ResNet, (l) Proposed SPCNet, and (m) Ground-reference map.

TABLE III

COMPUTATIONAL COSTS OF THE PROPOSED SPCNET WITH AND WITHOUT DOWNSAMPLING STRATEGIES

Downsampling strategies	Farmland		River		Hermiston		Bay Area	
	Parameters (K)	FLOPs (M)	Parameters (K)	FLOPs (M)	Parameters (K)	FLOPs (M)	Parameters (K)	FLOPs (M)
With	80.42	90.88	94.18	213.47	80.10	181.03	102.50	116.32
Without	121.12	137.77	145.89	332.61	120.55	274.22	160.87	183.56

TABLE IV

RESULTS OBTAINED BY THE PROPOSED SPCNET WITH DIFFERENT DOWNSAMPLING STRATEGIES (BOLD VALUES INDICATE THE MOST ACCURATE RESULTS FOR EACH VARIABLE AND EACH DATASET)

Downsampling strategies	Farmland			River			Hermiston			Bay Area		
	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)
Bilinear	97.63	94.24	95.91	96.97	**80.39**	**82.04**	96.41	89.67	91.98	98.20	96.39	98.31
Nearest Neighbor	97.50	93.93	95.69	**97.03**	80.38	82.00	96.43	89.66	91.95	98.24	96.47	98.35
Bicubic	**97.66**	**94.30**	**95.94**	97.01	80.29	81.91	**96.56**	**90.05**	**92.26**	**98.28**	**96.55**	**98.39**

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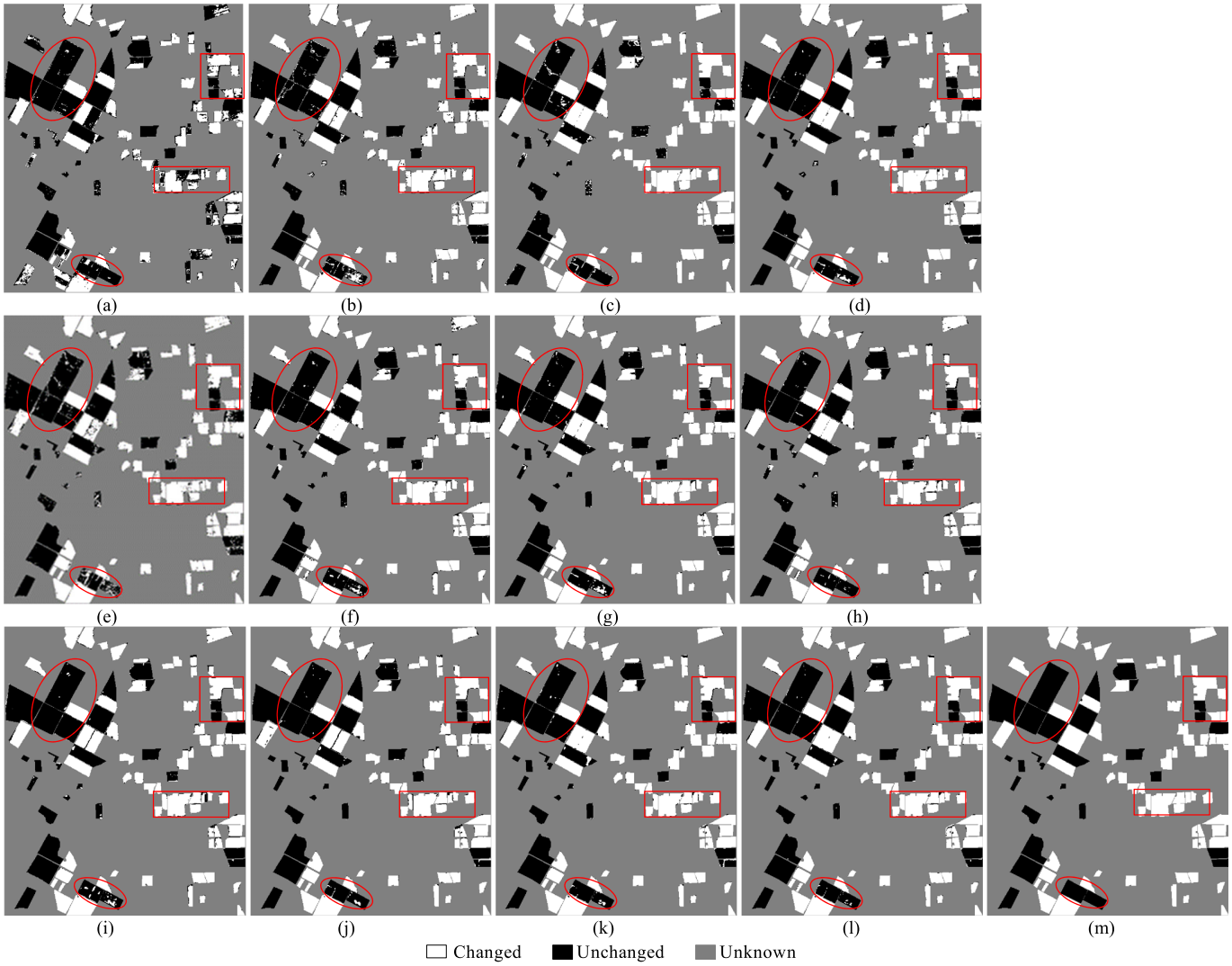


Fig. 9. CD maps produced by different methods on the Bay Area dataset: (a) CVA, (b) SVM, (c) SiamNet, (d) SSA-SiamNet, (e) GETNET, (f) Sub-2DCNN, (g) Con-2DCNN, (h) Con-3DCNN, (i) Sub-RSSAN, (j) Con-RSSAN, (k) ResNet, (l) Proposed SPCNet, and (m) Ground-reference map.

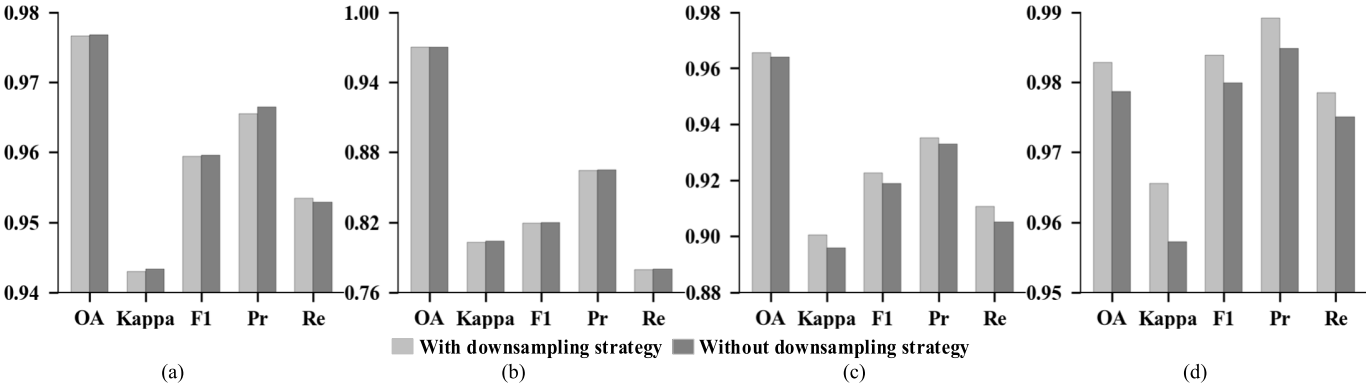


Fig. 10. Accuracy of the proposed SPCNet with and without downsampling strategies for (a) Farmland dataset, (b) River dataset, (c) Hermiston dataset, and (d) Bay Area dataset.

FLOPs with the downsampling operation are smaller than those without, which confirms the superiority of the proposed method with the downsampling operation.

2) Analysis of Different Downsampling Strategies: Different downsampling strategies, including bilinear, nearest neighbor, and bicubic, are considered to analyze their influence on the CD accuracy in the proposed SPCNet method. As shown in Table IV, the CD results with the three downsampling strategies are close in the four datasets. This finding illustrates the robustness of the proposed SPCNet to the downsampling strategy. Note that the bicubic strategy is employed in the proposed SPCNet method.

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TABLE V
RESULTS OBTAINED BY THE PROPOSED SPCNET WITH DIFFERENT POOLING STRATEGIES (BOLD VALUES INDICATE THE MOST ACCURATE RESULTS FOR EACH VARIABLE AND EACH DATASET)

Pooling strategies	Farmland			River			Hermiston			Bay Area		
	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)
Avg	97.56	94.05	95.77	96.89	79.96	81.66	96.45	89.75	92.03	98.30	96.58	98.40
Max	97.65	94.29	95.95	96.90	79.73	81.42	96.33	89.38	91.74	**98.38**	**96.74**	**98.48**
Avg+Max	**97.66**	94.30	95.94	97.01	80.29	81.91	**96.56**	**90.05**	**92.26**	98.28	96.55	98.39
[Avg, Max]	97.65	**94.31**	**95.96**	**97.03**	**80.85**	**82.48**	96.32	89.39	91.76	98.31	96.61	98.42

3) Analysis of the Pooling Strategy: We explored the influence of different pooling strategies on the CD results of the proposed SPC module. Four potential pooling strategies were considered in this article, which were the individual average-pooling operator (Avg), individual max-pooling operator (Max), sum of average- and max-pooling operators (Avg + Max), and concatenated average- and max-pooling operators ([Avg, Max]). The strides of MaxPool and AvgPool operators are set to 1. When different pooling strategies are used in the proposed SPCNet, all of the other configurations are kept fixed. As shown in Table V, the performances obtained with different pooling strategies are relatively close, which indicates that the SPC module is robust to the pooling strategy. Overall, Avg + Max is the most suitable for adoption in the proposed SPCNet.

IV. DISCUSSION AND CONCLUSION

In this article, we have proposed a general end-to-end SPCNet method to accomplish the CD task in bitemporal HSIs with different spatial resolutions. In the proposed SPCNet method, a modified ResNet is used as a baseline to detect land-cover changes. The subpixel convolution layer can fully extract the subpixel context information by learning an array of upscaling filters. Then, the designed SR block that contains an efficient subpixel convolution layer is introduced to upscale the LR feature maps into HR feature maps. Furthermore, the designed SPC module, which integrates the SR block and other related operators, is embedded into the LR branch to generate more discriminative representations. Compared with other benchmark methods, the improvement in terms of the CD accuracy obtained by the proposed SPCNet is mainly due to the application of the designed SPC module that can generate more discriminative representations of the subpixel context information. The proposed method represents a general end-to-end CD framework for bitemporal HSIs with different spatial resolutions.

As shown in Fig. 3(b), the HSIs acquired by the EO-1 satellite. (i.e., the Farmland, River, and Hermiston datasets) show a similar trend between the CD accuracy and the patch size. However, this trend is different for the Bay Area dataset acquired by AVIRIS. The apparent difference between the EO-1 satellite and AVIRIS is the spatial resolution. Therefore, it is worthwhile to further investigate the relation between increases in the CD results and the spatial resolution. The proposed SPCNet method has the potential to accomplish the CD task with other data sources, such as MSI, VHR images, and SAR images. In these cases, the hyperparameters, such as the kernel number, the batch size, and the patch size, should be properly determined. For example, unlike HSI, MSI and SAR images have few spectral bands, and thus, it is reasonable to reduce the kernel numbers. For VHR images, the patch size can be increased to acquire more correlation with the surrounding pixels.

The designed SPC module can be extended as a plug-and-play unit to other baseline networks and other HSI processing tasks, such as heterogeneous image fusion [43], hyperspectral unmixing [44], and subpixel mapping [45]. For different tasks, a key parameter is the scale factor for different resolution remote sensing images. In contrast, it would be interesting in future research to explore the relation between the number of SR blocks in the SPC module and the scale factor.

It is worthwhile to note that the experiments carried out on the proposed SPCNet method used synthetic LR images. This approach is a limitation of the current experimental setup. As a future development, we plan to test the proposed method on real bitemporal images that have different spatial resolutions. Moreover, we will explore the generalization of the method to multiple CD. In addition, the proposed SPCNet is a supervised method.

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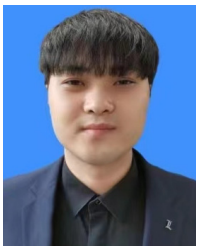
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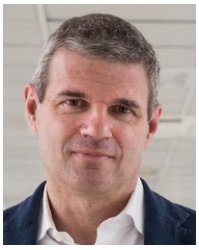
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